

NITTE (Deemed to be University)

(Established under Sec.(3) of UGC Act, 1956)

First Semester M.Tech (CBCS) Degree Examinations, December 2025

25MCT101 – DATA ANALYTICS

(For MCT)

Duration: 3 Hours]

[Max. Marks: 100

Note: Answer Five full questions choosing **One full** question from **each Module**.

Module – I

- | | Marks | BT* | CO* | PO* |
|--|-------|-----|-----|-----|
| 1. a) Find the inverse of $[A] = \begin{bmatrix} 2 & 3 & 4 \\ 4 & 3 & 1 \\ 1 & 2 & 4 \end{bmatrix}$ | 5 | 2 | 1 | 2 |
| b) $A = \begin{bmatrix} 8 & 0 & 0 \\ 4 & -2 & 0 \\ 3 & 6 & 0 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & -2 \\ 4 & 2 \\ -5 & 1 \end{bmatrix}$, then find the matrix X, such that $3A + 2X = 6B$. | 5 | 2 | 1 | 2 |
| c) Find QR decomposition of the matrix $[A] = \begin{bmatrix} 1 & 1 & 0 \\ 1 & -1 & 1 \\ 0 & 1 & 0 \end{bmatrix}$ | 10 | 2 | 1 | 2 |
| 2. a) If $A = \begin{bmatrix} 2 & 0 & 1 \\ 2 & 1 & 3 \\ 1 & -1 & 0 \end{bmatrix}$; Find $A^2 - 5A + 6I$ | 5 | 2 | 1 | 2 |
| b) Find the inverse of $[A] = \begin{bmatrix} 1 & 1 & 3 \\ 1 & 3 & -3 \\ -2 & -4 & -4 \end{bmatrix}$ | 5 | 2 | 1 | 2 |
| c) Solve by LU decomposition method
$x + 5y + z = 14$
$2x + y + 3z = 13$
$3x + y + 4z = 17$ | 10 | 2 | 1 | 2 |

Module – II

- | | | | | |
|--|----|---|---|---|
| 3. a) Explain the Bias-Variance Trade-off in the context of model complexity. How does it relate to overfitting and underfitting? | 5 | 1 | 2 | 1 |
| b) Why is logistic regression considered a linear model for classification, even though it predicts probabilities? | 5 | 1 | 2 | 1 |
| c) Explain the concept of Self-Supervised Learning (SSL).
(a) How does it differ from traditional supervised and unsupervised learning paradigms?
(b) Give a pretext task example for unlabeled text data. | 10 | 2 | 2 | 2 |
| 4. a) Contrast the goals of Dimensionality Reduction and Clustering in unsupervised learning. | 5 | 1 | 2 | 1 |
| b) What is the purpose of regularization in regression, and how do L1 and L2 regularization differ? | 5 | 1 | 2 | 1 |



- c) A PCA on vibration data gives eigenvalues [4.2, 1.1, 0.3, 0.1]. How many components would you retain to preserve 95% of variance?

10 2 2 2

Module – III

5. a) Solve the ODE by separating the variables:

$$y' = \sin^2(x-y+1)$$

6 2 3 2

- b) Solve the ODE using Leibnitz's linear equation:

$$\sec x \frac{dy}{dx} = y + \sin x$$

6 2 3 2

- c) Solve the higher order PDE and obtain complete solution:

$$(D^3 - 4D^2D' + 4DD'^2)z = 0$$

8 2 3 2

6. a) Solve using Leibnitz's linear equation:

$$x y' + 2 y - x \sin x = 0$$

6 2 3 2

- b) Solve by Lagrange method: $yzp - xzq = xy$

6 2 3 2

- c) Solve the higher order PDE:

$$r - 2s + t = \sin(2x+3y)$$

8 2 3 2

Module – IV

7. a) Find the Fourier series of

$$f(x) = \frac{\pi-x}{2} \text{ in an interval of } 0 \text{ to } 2\pi.$$

10 2 4 2

- b) Find Laplace Transform of $(\cosh at, \cos bt)$

10 2 4 2

8. a) Find the Fourier series of $f(x) = 2x - x^2$ in an interval of 0 to 3.

10 2 4 2

- b) Find Laplace Transform of $(t^2 - 3t + 2) \sin 3t$

10 2 4 2

Module – V

9. a) Discuss usefulness of statistics in Mechatronics.

8 1 5 1

- b) Data (n=5):

RPM: [1000, 1500, 2000, 2500, 3000]

Torque (Nm): [10, 15, 21, 25, 31]

Fit model $y = \beta_0 + \beta_1 x + \epsilon$. Compute $\hat{\beta}_0, \hat{\beta}_1$, SST, SSR, SSE, R^2 , standard error of slope, t-test for slope.

12 2 5 2

10. a) Explain Gibb's sampling taking a real word example.

8 1 5 1

- b) The specimen of copper wires drawn from a large lot have the following breaking strength (in kg. weight):

578, 572, 570, 568, 572, 578, 570, 572, 596, 544

Test (using Student's t-statistics) whether the mean breaking strength of the lot may be taken to be 578 kg. weight (Test at 5 percent level of significance).

12 2 5 2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
